

How Educational Differences are Reflected in Reddit Posts

Maxwell Champlin

3859119 / Maxc1102@gmail.com

Abstract

In this project we sought to understand the linguistic feature differences by education of reddit posters. As such we used 5000 posts from the r/PhD and r/Teenagers subreddit to be our main dataset for feature differences. As well, we wished to be able to classify posts by subreddit, for this we used 1000 posts. We sought to look at as many linguistic features as possible, as well as specialized educational tools like the Brysbert age of word acquisition and the Flesch-Kincaid reading ease and grade level scores. We found that the key differences between the two subreddits did not lay in general linguistic features, and that features like the amount of punctuation that we had expected to show higher levels of education found that teenagers used more than posters in PhD. Age of acquisition and reading grade level scores showed differences in the direction we had expected, however, the reading level and age of word acquisition of posts for both subreddits were well below our expectations. For our classifier we found that classifying the subreddits was relatively easy when looking at the total samples, as the words used between them were often very different. These results highlight the fact that expectations of educational differences showing up in the linguistic features of text is not likely, at least on informal sources, and that education is best found by looking at lexical selection.

1 Introduction

Language is an innate part of the human cognition system. Perhaps it is through Chomsky's proposed genetic faculty for language, or perhaps not. What we can be certain of is that reading and writing are not natural skills that humans are imbued with, instead they are skills that take many years to learn and are require artificial constructions like spelling and punctuation to convey an intended meaning, this is not a crutch that spoken language needs. Using subreddit data from r/Teenagers and r/PhD, we

hope to be able to examine the linguistic differences between the two groups and be able to examine how length of time in education plays a roll in various linguistic features. In particular we will be examining the overall reading grade level and reading ease of the posts using the Flesch-Kincaid reading ease and grade level calculations. This will allow us to get an overview of how these two communities differ lexically. We will also be using a number of features, like the amount of punctuation, number of unique nouns per post, and more to see how the communities vary and if our predictions of more formal tools like punctuation being something that is mostly used by those with more education.

2 Data

In this project the dataset used came from two subreddits: r/Teenagers and r/PhD. The data used was a subsample of 5000 posts, which was then filtered to only include those over 10 and less than 500 words. In total we had about 2696 posts for the Teenager subreddit and about 3286 for the PhD subreddit. The choice for 5000 posts which was then filtered was made considering the statistical significance of our set over the 1 million or so posts in the teenagers dataset. As well, the size was chosen to reflect the limitations on computing power as each additional post added about a second of processing time for the machine.

For the classifier we created, we used 10,000 posts as our dataset. This was due to the higher demand for relevant data that was needed for the classification tasks than for the linguistic feature identification. While we have chosen to have two separate datasets, the data from both is reliable as there is more than sufficient data to have an accurate sampling of the data at hand.

In the PhD subreddit our median number of words (post filtering) per post is 78, this is higher than the teenagers who are at

3 Features

Describe the set of features you are going to use. Give examples for features that are a bit less intuitive to interpret. Repeat a hypothesis that you have regarding the relationship between certain features and certain empathy types or how they differ between communities or speakers.

For the research questions we will use a mixture of pre-built and manual features from LFTK to analyze the data. In the area of how educational differences show up in the posts of r/PhD posters or r/Teenager posters we looked at the prebuilt LFTK Flesch-Kincaid Grade Levels¹, the Flesch-Kincaid Reading Ease², and the Brysbert age of acquisition of words, and of sentences³.

The specialized metrics for grade level and reading ease were chosen due to them offering a clear measurement of lexical differences between the two subreddits. Particularly the age of acquisition metrics offer a clear look at the lexical differences with a more robust methodology than the reading ease and grade level which, while useful, only concern themselves with the number of syllables per word, and could be less accurate than expected due to proper nouns or cases where words are not properly spaced. The use of the reading ease and grade level statistics is that they allow us to see the overall differences in word length, which while not a perfect representation of lexical choices allows us to see a rough idea of what the overall lexical differences are by word length.

As well, as LFTK is a powerful tool, we are also looking at a full suite of syntactic features in the data. The key features that we intend to look at based on our hypotheses of what linguistic features correlate with education levels, we expect that PhD will have more punctuation, as that is often associated with more formal writing. As well we expect the number of unique nouns to be higher for PhD as we expect that more educated people will use more unique nouns than teenagers will. Since we have the ability to look at a large number

¹The Grade level used are American Grade Levels. The score is calculated by dividing the total words by total sentences and adding it to the total syllables over total words. The exact equation can be found [Here](#)

²The Reading Ease score is calculated by dividing the total words by total sentences, and then minus the total syllables over total words. There are a number of scalars in the calculation to be able to get the 0-100 scale. Further information [Here](#)

³The age of acquisition is determined through surveys and currently contains about 44,000 words.

of features, we will report later surprise results from our feature variation examination.

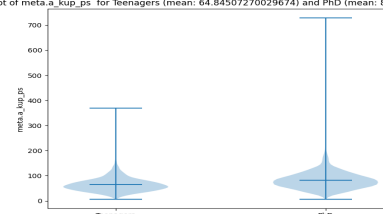
4 Methodology

Describe the steps you are going to perform in order to retrieve results. (1) pre-processing (do you convert the dataset to a Corpus Object? Do you do any cleaning of the data?) (2) feature extraction: which tool(s) do you use? (3) Experimental setup for prediction task: What are your data splits? Best practice for all experiments (now and in the future): always use several (different) folds (at least 3) for training + evaluation. Report the average and standard deviation of performance results. If your target label is imbalanced, report F1 Macro Score and not accuracy. Also report the result for the positive and negative class (so you know whether you are overpredicting or undergeneralizing). (4) For the feature analysis (either correlation with empathy or variation between communities), what methods are you using? E.g. plots, statistical model, or maybe you are calculating correlations? Identifying discriminating words based on frequency? Feature importance in the classification?

5 Results

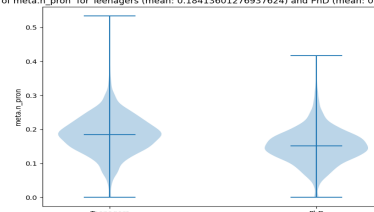
Here you should be able to answer the research questions you formulated in the beginning.

Violin Plot of meta_a_kup_ps for Teenagers (mean: 64.84507270029674) and PhD (mean: 81.65951095556908)



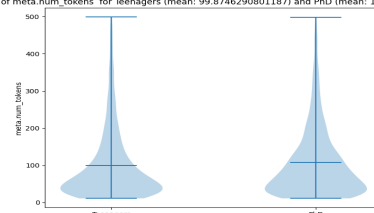
Average Kuperman age of acquisition per sentence

Violin Plot of meta_n_pron for Teenagers (mean: 0.18413601276937624) and PhD (mean: 0.1513662807787332)

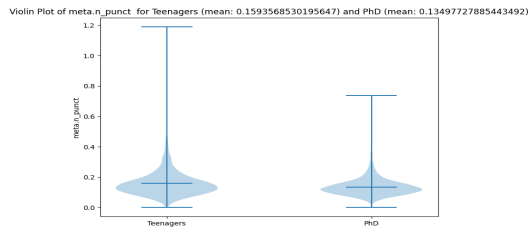


number of pronouns

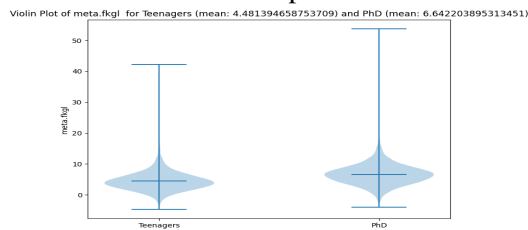
Violin Plot of meta_num_tokens for Teenagers (mean: 99.8746290801187) and PhD (mean: 107.24528301886792)



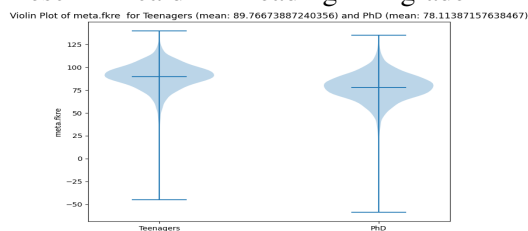
number of tokens



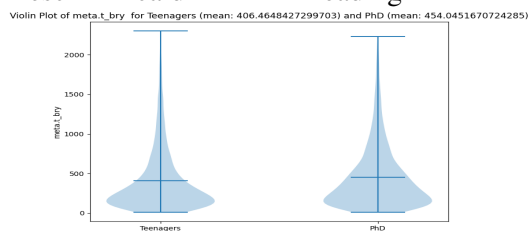
number of punctuation marks



Flesch-Kincaid reading grade level



Flesch-Kincaid reading Ease



Total bryant age of acquisition of words.

5.1 Predicting Empathy / Predicting source community

What are the results of the classification task? (Table and main findings) If you do empathy, report how many instances were correctly predicted (out of the 20).

5.2 Linguistic Features

What are the results with respect to specific features? Plots, ranking of important features, examples, relevant words etc.

6 Discussion

Reflect on your findings: did you see expected results? Did you see something unexpected? What data points were missed or miss-classified by the model(s) and why? What other features would be promising? What additional information would be useful to know or annotate to perform better? Did you find particular properties of the data that

made classification more or less reliable (e.g. reddit posts are over a long time span, the way empathy is annotated is oversimplified ...)

7 Conclusion

Summarize your main findings with respect to the research questions and your main discussion points / biggest problem for future work.

A Example Appendix

A.1 Use of generative AI

If you used generative AI for the allowed purposes, specify here what for and which one you used.

A.2 Additional Figures and Tables